

There are four disks in a "RAID-5" unit, which requires that corresponding bits on 1/1 the four disks sum to 0 (modulo 2). We shall focus on four strings of five bits each. The four strings are chosen from corresponding bit positions on the four disks. The strings from the first two disks are 01100 and 10101. First, determine from the given information what constraints there are on the values of the strings on the third and fourth disks. Then, identify from the list below, which of the following could not be the strings from the third and fourth disks.

☒ 10101 and 00111

☒ 10001 and 00001

☐ 10110 and 01111

☒ 10011 and 11010

Suppose we use double buffering to read blocks from a hard disk to improve the 1/1 IO performance. That is, the disk controller has two buffers: one to store the data from a disk block, and the other for an application process to consume the data of another block. Assume each disk block needs time R to read, and the application needs time P to process one block. What is the total time for the application to read in 10 disk blocks and consume the data?

☐ No matter what R and P are, the total time is always $10 * P + R$.

☐ If $R = P$, then the total time is $10 * R$.

☐ If $R > P$, then the total time is $10 * P + R$.

☒ If $P = 15\text{ms}$, and $R = 9\text{ms}$, then the total time is 159.

Relation $R(A,B,C)$ currently has only the tuple $(0,0,0)$, and it must always satisfy the functional dependencies $A \rightarrow B$ and $B \rightarrow C$. What condition(s) must be obeyed by any tuple that may be inserted without violating either of these FD's? Identify from the list below the tuples that cannot be inserted into R legally without violating the FD constraints.

☒ $(0, 0, 2)$

☒ $(1, 0, 2)$

☒ $(0, 1, 0)$

☐ $(1, 0, 0)$

Let $R(ABCDEFGH)$ satisfy the following functional dependencies: $A \rightarrow B$, $CH \rightarrow A$, $B \rightarrow E$, $BD \rightarrow C$, $EG \rightarrow H$, $DE \rightarrow F$. Which of the following FD's are not satisfied by R ?

☒ $BFG \rightarrow AE$

☒ $ADE \rightarrow CH$

☒ $ACG \rightarrow DH$

☐ $BDG \rightarrow AE$

Consider relation $R(A,B,C,D,E)$ with functional dependencies $AB \rightarrow C$, $BC \rightarrow D$, $CD \rightarrow E$, $DE \rightarrow A$, and $AE \rightarrow B$. If we project R onto the relation $S(A,B,C,D)$, that is, $S = \pi_{A,B,C,D}(R)$, then certain functional dependencies must hold in S . These are the FD's that can be inferred from those of R , but that involve only the attributes of S . Note that the inference itself is not restricted only to attributes of S , just the result. Which of the following FD's do not hold in S ?

☒ $A \twoheadrightarrow D$

☐ $ABC \twoheadrightarrow D$

☒ $AE \twoheadrightarrow B$

☒ $C \twoheadrightarrow B$